University of Florida, Department of Mathematics

Analysis First-Year Exam, May 2019, Part 2

Do exactly 4 problems. Work must be presented in a neat and logical fashion in order to receive credit. Do not leave any gaps. When a theorem is used in a proof it must be precisely stated.

1. Let $f_n : [a,b] \to \mathbb{R}$ be a uniformly bounded sequence of Riemann integrable functions. Prove that if $f_n \to f$ uniformly, then f is Riemann integrable.

2. Let $f : [0,1] \to \mathbb{R}$ be continuous. Suppose that

$$\int_0^1 f(x)x^{2019k}\,dx = 0$$

for all integers $k \geq 0$. Must f be identically 0? Prove, or give a counterexample.

3. Let $f_n : [0,1] \to \mathbb{R}$ be a sequence of measurable functions and suppose there is a function $g \in L^1[0,1]$ such that $|f_n| \leq g$ for all n. Consider the functions

$$F_n(t) = \int_0^t f_n(x)\,dx.$$

Prove

i) each F_n is continuous on $[0,1]$

ii) there is a subsequence (F_{n_k}) converging uniformly on $[0,1]$.

4. Let $(X, \mathcal{M})$ be a measurable space and $f_n : X \to \mathbb{R}$, $n = 1, 2, 3, \ldots$, a sequence of measurable functions. Prove that each of the following subsets of X is measurable:

a) $\{x \mid f_n(x) > 0 \text{ for infinitely many values of } n\}$

b) $\{x \mid \text{the sequence } (f_n(x)) \text{ is eventually monotone}\}$

c) $\left\{x \,\middle|\, \lim_{n\to\infty} nf_n(x) = 0\right\}$

5. Let $f_1 \geq f_2 \geq f_3 \geq \cdots$ be nonnegative measurable functions on a measure space $(X, \mathcal{M}, \mu)$, and put $f = \lim f_n$. Suppose that $\int f_k\,d\mu < \infty$ for some k. Prove that

$$\int f\,d\mu = \lim \int f_n\,d\mu.$$

Give a counterexample to show that the conclusion can fail if the finiteness hypothesis is dropped.